MATH4321

Lecture 1: Static games of complete information and Nash equilibrium

1.1 Static Games

single (once-and-for-all) decision simultaneously and independently; outcome and payoffs; Complete information

1.1.1 Games in normal form

Definition (Games in normal form)**.** A game in normal form is a games consisting of the following 3 elements:
(i) A finite set of players, denoted by $P = \{1, 2, \ldots, n\}$;
(ii) A set of actions (can be finite or infinite) that can be chosen by player i, denoted by S_i. S_i is called strategic set of player i. Each player chooses its strategy $a_a \in S_i$ simultaneously and independently (without the influence from other players);
(iii) Payoff function of each player, denoted by $V_i(a_1, a_2, \ldots, a_n)$ or $V_i(a_i, a_{-i})$, describing the payoff received by player i given the strategies chosen by the players. Here, $a_{-i} = (a_1, a_2, \ldots, a_{i-1}, a_{i+1}, \ldots, a_n)$ denotes actions taken by other players. Mathematically, the games in normal form can be described as an order triple $G = (P, \{S_i\}_{i \in P}, \{V_i\}_{i \in P})$.

e.g. for a 2-player Paper-Scissor-rock games, its normal form can be expressed as:
1. Set of players: $N = \{1, 2\}$;
2. Strategic set of player i: $S_i = \{\text{Paper}, \text{Scissor}, \text{Rock}\}$ (Denoted by P, S, R respectively);
3. Payoff function: $V_i(R, R) = V_i(S, S) = V_i(P, P) = 0$, $V_i(R, S) = V_i(S, P) = V_i(P, R) = 1$,
$V_i(R, P) = V_i(S, R) = V_i(P, S) = -1$
The payoff functions can also be expressed as:

	Paper	*Scissor*	*Rock*
Paper	$(0,0)$	$(-1,1)$	$(1,-1)$
Scissor	$(1,-1)$	$(0,0)$	$(-1,1)$
Rock	$(-1,1)$	$(1,-1)$	$(0,0)$

Such games is sometimes called matrix games.

1.2 equilibrium(pure strategy)

Premise: All players are rational; All players are intelligent; All players should engage in non-cooperative behaviour.
We want to predict the strategy adopted by other players in order to determine our own best strategy. We have 2 ways of achieveing this:
1. Eliminating dominated strategies - Rule out all strategies which the rivals will not choose.
2. Best response and Nash equilibrium - Predict what will rivals do based on their payoff functions.

1.2.1 Dominated strategy

We first assume that each player adopts pure strategy(strategy that a single player chooses a single action a_i from his/her strategic set S_i).
If the payoff of strategy A is always worse-off than strategy B for player 1, then we say that strategy A is strictly dominated by strategy B for player 1, and the strategy A is called dominated strategy.

Definition (Dominated strategy for pure strategy)**.** We let $s_i \in S_i$ and $s_i' \in S_i$ be two possible strategies for player i. We say that s_i is strictly dominated by the strategy s_i' if for any strategies adopted by other players (denoted by $s_{-i} = (s_1, s_2, \ldots, s_{i-1}, s_{i+1}, \ldots, s_n)$), one has

$$V_i(s_i; s_{-i}) < V_i(s_i'; s_{-i})$$

That is, player i is always worse-off by adopting the strategy s_i than s_i', regardless of the strategies adopted by other players. We say $s_i \in S_i$ is a dominated strategy for player i if there is another strategy $s_i' \in S_i$ such that s_i is strictly dominated by s_i'.

Then we can use dominated strategy to simplify the solution procedure: e.g.

	Player 2		
Player 1	A	B	C
A	$(4, \mathbf{2})$	$(1, \mathbf{5})$	$(5, 4)$
B	$(3, \mathbf{3})$	$(1, \mathbf{4})$	$(3, 2)$
C	$(2, \mathbf{4})$	$(6, \mathbf{5})$	$(6, 7)$

Then we can rule out strategy A of player 2.

1.2.2 Iterative Elimination of Strictly Dominated Strategies (IESDS)

We first consider player 1 and her strategic set S_1. We remove all dominated strategies from her strategic set. We then repeat this process for player 2, 3, .. , n sequentially. Let $S_i^{(k)}$ denote the new strategic set of player i after k rounds of eliminations.
Repeat the elimination process until $S_i^{(k+1)} = S_i^{(k)}$ for all i. That is, there is no more dominated strategy identified.

Definition (Iterated-elimination equilibrium)**.** A strategic profile $s = (s_1, s_2, \ldots, s_n)$ is said to be iterated-eliminated equilibrium if it survives the process of IESDS.
Attention : it must be strictly dominated. ONLY $<$, NO $\leq$.

e.g. Cournot Duopoly:
There are two identical firms (Firm 1 and Firm 2) producing some products for a same market. Two firms need to decide the number of products (denoted by $q_i \in [0, \infty), i = 1, 2$) produced. It is given that
1.The total cost of production of firm i is $c(q_i) = 10q_i$.
2.The market price (selling price) of the product, which is driven by supply and demand, is assumed to be $p(q_1, q_2) = 100 - q_1 - q_2$.
Suppose that two firms choose to produce q_1 and q_2 units of goods respectively, the profits of the two firms. denoted by $V_1(q_1, q_2)$ and $V_2(q_1, q_2)$ respectively, can be expressed as
$V_1(q_1, q_2) = \underbrace{q_1(100 - q_1 - q_2)}_{\text{revenue = price} \times \text{quantity}} - \underbrace{10q_1}_{\text{cost}} = 90q_1 - q_1^2 - q_1 q_2;$

$V_2(q_1, q_2) = q_2(100 - q_1 - q_2) - 10q_2 = 90q_2 - q_2^2 - q_1q_2$.
By applying IESDS, find the final strategies adopted by each firm.
Solution:
Step1:
We first consider firm 1 and rule out dominated strategy from his strategic set. Intuitively, one should expect that it is always sub-optimal for the firms to produce too many products (q_1 is too large) or too few products (q_1 is too small). To see this, we note that

$$\frac{\partial}{\partial q_1}V_1(q_1, q_2) = \underbrace{90 - 2q_1}_{<0 \text{ for } q_i > 45} - q_2 \ldots (1)$$

It is clear that $\frac{\partial V_1(q_1,q_2)}{\partial q_1} < 0$ for $q_1 > 45$ and $q_2 \geq 0$. This implies the firm 1's payoff will decrease if it chooses to increase q_i beyond 45. Using this fact, one can deduce that for any $q_1 > 45$,

$$\underbrace{V_1(q_1, q_2)}_{\text{payoff (choose } q_1 > 45)} < \underbrace{V_1(45, q_2)}_{\text{payoff (choose } q_1 = 45)}, \quad \text{for any } q_2 \geq 0.$$

Since any $q_1 > 45$ is strictly dominated by the strategy $q_1 = 45$, so $q_1 > 45$ is a dominated strategy for player 1. Similarly, $q_2 > 45$ is a dominated strategy for firm 2. Both strategic sets are narrowed down to $S_1^{(1)} = S_2^{(1)} = [0, 45]$.

Step2:
Given the new strategic set, we proceed to identify more dominated strategies. We consider player 1 again. Using the fact that $q_2 \leq 45$, one can observe that

$$\frac{\partial}{\partial q_1}V_1(q_1, q_2) = 90 - 2q_1 - q_2 \geq 90 - 2q_1 - 45 = \underbrace{45 - 2q_1}_{>0 \text{ for } q_1 < 22.5}.$$

We observe that $\frac{\partial}{\partial q_1}V_1(q_1, q_2) > 0$ for $q_1 < 22.5$. This implies that

$$\underbrace{V_1(q_1, q_2)}_{\text{payoff (choose } q_1 < 22.5)} < \underbrace{V_1(22.5, q_2)}_{\text{payoff (choose } q_1 = 22.5)} \quad \text{for any } q_1 < 22.5.$$

Therefore, $q_1 < 22.5$ is also a dominated strategy for firm 1. Similarly, one can deduce that $q_2 < 22.5$ is a dominated strategy for firm 2. After the second stage of elimination, the strategy sets of both firms are narrowed down to

$$S_1^{(2)} = S_2^{(2)} = [22.5, 45].$$

Remark: One may further reduce the strategic set using $q_i \geq 0$ (since $S_i^{(1)} = [0, 45]$). However, this only implies that $q_i > 45$ is a dominated strategy (same as Step 1).
Step3:
We let $S_i^{(k)} = [a_k, b_k]$ be the strategic set of player i after k-th round of elimination. We consider the $(k+1)^{th}$ round of the elimination process:

Since $q_j \geq a_k$, we can deduce that

$$\frac{\partial}{\partial q_i}V_i(q_i, q_j) = 90 - 2q_i - q_j \leq 90 - 2q_i - a_k.$$

This implies that $\frac{\partial V_i}{\partial q_i} < 0$ when $q_i > \frac{90-a_k}{2}$. We get

$$V_i(q_i; q_j) < V_i\left(\frac{90 - a_k}{2}; q_j\right) \quad \text{for } q_i > \frac{90 - a_k}{2} \text{ and } q_j \geq a_k.$$

So any $q_i > \frac{90-a_k}{2}$ is a dominated strategy for player i.

Similarly, using the fact that $q_j \leq b_k$, we obtain

$$\frac{\partial}{\partial q_i} V_i(q_i, q_j) = 90 - 2q_i - q_j \geq 90 - 2q_i - b_k.$$

This implies that $\frac{\partial V_i}{\partial q_i} > 0$ when $q_i < \frac{90-b_k}{2}$. We get

$$V_i(q_i; q_j) < V_i\left(\frac{90 - b_k}{2}; q_j\right) \quad \text{for } q_i < \frac{90 - b_k}{2} \text{ and } q_j \leq b_k.$$

So any $q_i < \frac{90-b_k}{2}$ is a dominated strategy for player i.

Combining the results, we deduce that $S_i^{(k+1)} = [a_{k+1}, b_{k+1}]$ where

$$a_{k+1} = \frac{90 - b_k}{2}, \quad b_{k+1} = \frac{90 - a_k}{2}.$$

By continuing the elimination process, one can narrow down the strategy sets continuously and we summarize the result in the following table:

k	$S_1^{(k)} = S_2^{(k)}$
1	$[0, 45]$
2	$[22.5, 45]$
3	$[22.5, 33.75]$
4	$[28.125, 33.75]$
5	$[28.125, 30.9375]$
6	$[29.53125, 30.9375]$
7	$[29.53125, 30.23438]$
$\vdots$	$\vdots$
∞	$[30, 30]$

We observe from the above table that the upper bound and lower bound of the strategic set are getting close to 30. In fact, one can show that the strategic sets will become a singleton. Thus, we conclude that both firms will produce 30 units of products respectively in equilibrium.
Proof: using the recursive relation derived in Step 3. We summarize the key steps below and omit the detailed verification.

Recall that $S_i^{(1)} = [0, 45]$ and $S_i^{(2)} = [22.5, 45]$ from Step 1 and 2. We take $a_1 = 0$, $a_2 = 22.5$, $b_1 = b_2 = 45$.
Step 1: Argue that $22.5 \leq a_k \leq b_k \leq 45$ for $k \geq 2$. This can be done using mathematical induction.
Step 2: Argue that the sequence $\{a_k\}$ is increasing and $\{b_k\}$ is decreasing, i.e., $a_k \leq a_{k+1} \leq b_{k+1} \leq b_k$, using mathematical induction.
Step 3: Combining the above results, one can deduce that: $S_i^{(k+1)} \subseteq S_i^{(k)} \subseteq [0, 45]$
Both sequences $\{a_1, a_2, \dots\}$ and $\{b_1, b_2, \dots\}$ converge since the sequence $\{a_k\}$ is increasing and bounded from above, and the sequence $\{b_k\}$ is decreasing and bounded from below.
Step 4: We let $\lim_{k\to\infty} a_k = a$ and $\lim_{k\to\infty} b_k = b$. By taking limits on the recursive relations, one can show that $a = b = 30$. So that $S_i^{(\infty)} = \bigcap_{k=1}^{\infty} S_i^{(k)} = [a, b] = [30, 30]$, which is a singleton. Thus, we conclude that both firms will produce 30 units of products in equilibrium.

Mind that IESDS may not lead to unique solution. It is because the scheme only rule out all strategies that are obviously (or always) sub-optimal for the players.

1.2.3 Best response and Nash equilibrium

In a n-person games, a player needs to figure out all possible strategies that will be adopted by other players in order to determine her own optimal strategy. We call such prediction as belief (denoted by B_i) of the player.

Definition (Belief)**.** In an n-person game, a player's belief B_i is a subset of

$$S_{-i} = S_1 \times S_2 \times \cdots \times S_{i-1} \times S_{i+1} \times \cdots \times S_n$$

that collects a set of possible strategies adopted by other players.

Initially, a player's belief will be the entire strategic sets of other players. Since all players are rational and intelligent, the belief can be sharpen by considering the optimality of his opponents. One possible way is to eliminate all strategies that won't be chosen by the opponents such as dominated strategy.

Definition (Best response)**.** Given a set of opponents' strategy $s_{-i} \in S_{-i}$, a player i's strategy $s_i^* \in S_i$ is said to be a **best response** to the strategy s_{-i} if and only if the corresponding payoff of player i is maximized over his strategic set:

$$V_i(s_i^*; s_{-i}) = \max_{s_i \in S_i} V_i(s_i; s_{-i}).$$

Here, the collection of best responses (with respect to s_{-i}) is denoted by $BR_i(s_{-i})$.

Remark:
1. The best response to the given opponents' strategies s_{-i} may not be unique.
2. The best response depends on the rival's strategy.

A profile $s^* = (s_1^*, s_2^*, \ldots, s_n^*)$ is the desired outcome of the games only when player i has made correct prediction on opponents' strategies and s_i^* is player i's best response to opponents' strategies.

Definition (Nash equilibrium for pure strategy)**.** A strategic profile $\mathbf{s}^* = (s_1^*, s_2^*, \ldots, s_n^*)$ is called a **pure strategy Nash equilibrium** if and only if for any player i:

$$V_i(s_i^*; s_{-i}^*) = \max_{s \in S_i} V_i(s; s_{-i}^*) \geq V_i(s_i; s_{-i}^*), \quad \text{for any } s_i \in S_i \ldots \quad (***)$$

In other words, no player has a strict incentive to deviate from the strategy s_i^* and adopt other strategies.

e.g. (Cournot Duopoly)
player 1's best response:

$$\left.\frac{\partial V_1(q_1, q_2)}{\partial q_1}\right|_{q_1 = q_1^*(q_2)} = 0 \quad \Rightarrow \quad \left.(90 - 2q_1 - q_2)\right|_{q_1 = q_1^*(q_2)} = 0$$

$$\Rightarrow \quad q_1^*(q_2) = \frac{90 - q_2}{2} \qquad (1)$$

Since $\left.\frac{\partial^2 V_1(q_1,q_2)}{\partial q_1^2}\right|_{q_1 = q_1^*(q_2)} = -2 < 0$, player 1's payoff is maximized when $q_1 = q_1^*(q_2)$, and $q_1^* = q_1^*(q_2)$ is the best response of player 1.
Similarly, we can get player 2's best response $q_2^*(q_1) = \frac{90-q_1}{2}$ (2).
With (1) and (2), we get $(q_1^*, q_2^*) = (30, 30)$ is the (unique) Nash equilibrium of the games.

Remark:
1. pure-strategy Nash equilibrium does not always exist.
2. Any dominated strategy is never a best response.
3. A strategic profile $s^* = (s_1^*, s_2^*, \ldots, s_n^*)$ uniquely survives IESDS in a **finite** normal-form game(finite players, finite strategies), then s^* is Nash equilibrium.
(Proof of 3: Assume the opposite as not equilibrium, then for any s^*, always have $s^{(1)}, s^{(2)}, \ldots, s^{(n)}$ such that $s^{(k)}$ have higher payoff than s^*, and since n is arbitrary, we can get infinite strategies $s^{(k)}$ from this process, which is contradicted to "finite" in the assumption.)

1.2.4 equilibrium refinement

Equilibrium refinement is the process to impose some addition conditions to rule out the equilibria that are unlikely to occur, including:
1. Pareto-dominance: Equilibrium selection with respect to the players' payoff.
2. Risk-dominance: Equilibrium selection with respect to the "stability" of the equilibrium.

Definition (Pareto-dominance and Pareto-optimality)**.** In an n-person game, we say a strategic profile (or equilibrium) $\mathbf{s} = (s_1, s_2, \ldots, s_n)$ **Pareto dominates** another strategic profile $s' = (s'_1, s'_2, \ldots, s'_n)$ if and only if

$$V_i(s_i; s_{-i}) \geq V_i(s'_i; s'_{-i}), \quad \text{for all } i = 1, 2, \ldots, n, \quad (1)$$

and

$$V_i(s_i; s_{-i}) > V_i(s'_i; s'_{-i}), \quad \text{for some } i = 1, 2, \ldots, n. \quad (2)$$

On the other hand, we say a strategic profile $\mathbf{s}^* = (s_1^*, s_2^*, \ldots, s_n^*)$ is **Pareto optimal** if there is no strategy $\mathbf{s}'(\neq \mathbf{s}^*)$ that Pareto dominates it.

Remark:
1. (1): none of the players will be worse off if all players chooses the strategy s instead of s'.
2. (2): at least one of the players will be better off if all players chooses the strategy s instead of s'.
3. the Pareto optimal equilibria is not necessarily to be unique. (This may happen: any equilibrium is not Pareto-dominated by any other equilibrium.)

Risk Dominance: Given a pair of Nash equilibria $[(s_1^*, s_2^*)$ and $(s_1^{**}, s_2^{**})]$. To compare the stabilities, for the Nash equilibrium (s_1^*, s_2^*), we define the **deviation cost** (the product of losses if choose to play another strategy) as:

$$\Phi(s_1^*, s_2^*) = \underbrace{\left(V_1(s_1^*; s_2^*) - V_1(s_1^{**}; s_2^*)\right)}_{\substack{\text{player 1's loss} \\ \text{(choose } s_1^{**} \text{ instead of } s_1^*)}} \underbrace{\left(V_2(s_2^*; s_1^*) - V_2(s_2^{**}; s_1^*)\right)}_{\substack{\text{player 2's loss} \\ \text{(choose } s_2^{**} \text{ instead of } s_2^*)}}.$$

Similarly, one can define the corresponding deviation cost for (s_1^{**}, s_2^{**}). That is,

$$\Phi(s_1^{**}, s_2^{**}) = \underbrace{\left(V_1(s_1^{**}; s_2^{**}) - V_1(s_1^*; s_2^{**})\right)}_{\substack{\text{player 1's loss} \\ \text{(choose } s_1^* \text{ instead of } s_1^{**})}} \underbrace{\left(V_2(s_2^{**}; s_1^{**}) - V_2(s_2^*; s_1^{**})\right)}_{\substack{\text{player 2's loss} \\ \text{(choose } s_2^* \text{ instead of } s_2^{**})}}.$$

Attention: when calculating $V_2(s_2^*; s_1^{**})$, read the matrix in reverse order.

Definition (Risk-dominance, Harsanyi and Selten (1988))**.** Let (s_1^*, s_2^*) and (s_1^{**}, s_2^{**}) be two pure strategy Nash equilibria in a 2-person game. (s_1^*, s_2^*) is said to be **risk-dominant** against another equilibrium (s_1^{**}, s_2^{**}) if and only if

$$\Phi(s_1^*, s_2^*) > \Phi(s_1^{**}, s_2^{**}).$$

Remark:
1. Higher deviation cost $\Longrightarrow$ less incentive to deviate $\Longrightarrow$ the equilibrium is more stable.
2. Higher deviation cost $\Longrightarrow$ risk-dominant

1.3 Mixed strategy

Definition (Mixed strategy for finite strategic set)**.** Let $S_i = \{a_{i1}, a_{i2}, \ldots, a_{iN_i}\}$ be the strategy set of player i. A **mixed strategy** of player i is a vector $\sigma_i = \big(\sigma_i(a_{i1}), \sigma_i(a_{i2}), \ldots, \sigma_i(a_{iN_i})\big)$ that describes a probability distribution over the strategy set S_i, where $\sigma_i(a_{ij}) = P(s_i = a_{ij})$ denotes the probability that player i chooses action a_{ij}. (Note: N_i denotes the number of actions in S_i.)

Here, the probability $\sigma_i(a_{ij})$ must satisfy the following properties:

- $\sigma_i(a_{ij}) \geq 0$, and
- $\displaystyle\sum_{j=1}^{N_i} \sigma_i(a_{ij}) = 1$.

Player's payoff: Expected payoff

Given the mixed strategies $\sigma_1, \sigma_2, \ldots, \sigma_n$ chosen by players, player i's expected payoff, denoted by $V_i(\sigma_i; \sigma_{-i})$, can be expressed as

$V_i(\sigma_i;\sigma_{-i}) = \sum_{\vec{a}\in S_1\times\cdots\times S_n} P\big(\text{players choose } \vec{a} = (a_1, a_2, \ldots, a_n)\big)\, V_i(a_i; a_{-i})$

$= \sum_{\vec{a}\in S_1\times\cdots\times S_n} \Big(\prod_{k=1}^{n} P\big(\text{player } k \text{ chooses } a_k\big)\Big)\, V_i(a_i;a_{-i}) = \sum_{\vec{a}\in S_1\times\cdots\times S_n} \Big(\prod_{k=1}^{n} \sigma_k(a_k)\Big)\, V_i(a_i;a_{-i})$

$= \sum_{a_i\in S_i}\sum_{a_{-i}\in S_{-i}} \sigma_i(a_i)\Big(\prod_{\substack{k=1,\ldots,n\\ k\neq i}} \sigma_k(a_k)\Big)\, V_i(a_i;a_{-i}) = \sum_{a_i\in S_i}\sum_{a_{-i}\in S_{-i}} \sigma_i(a_i)\, \underbrace{\Big(\prod_{\substack{k=1,\ldots,n\\ k\neq i}} \sigma_k(a_k)\Big)}_{\text{denoted by } \sigma_{-i}(a_{-i})}\, V_i(a_i;a_{-i})$

$= \sum_{a_i\in S_i} \sigma_i(a_i)\, \underbrace{\Big(\sum_{a_{-i}\in S_{-i}} \sigma_{-i}(a_{-i}) V_i(a_i;a_{-i})\Big)}_{V_i(a_i;\sigma_{-i})}.$

The second equality follows from the assumption that each player chooses her strategy independently.

1.3.1 Nash equilibrium for mixed strategy

Definition (Nash equilibrium for mixed strategy)**.** The strategic profile $\sigma = (\sigma_1^*, \sigma_2^*, \ldots, \sigma_n^*)$ is a Nash equilibrium if and only if for any player i,

$$V_i(\sigma_i^*;\sigma_{-i}^*) \geq V_i(\sigma_i;\sigma_{-i}^*),$$

for any mixed strategy σ_i of player i.

e.g. We consider the game with the following payoff matrix:

	Sushi (S)	Buffet (B)
Sushi (S)	$(3,2)$	$(0,1)$
Buffet (B)	$(0,0)$	$(2,3)$

Find all possible mixed-strategy Nash equilibria of this game.

Solution

Let $\sigma_1 = (p, 1-p)$ and $\sigma_2 = (q, 1-q)$ be the mixed strategies of the two players. The expected payoffs are

$$\begin{aligned} V_1(p;q) &= pq(3) + p(1-q)(0) + (1-p)q(0) + (1-p)(1-q)(2) \\ &= 5pq - 2p - 2q + 2, \\ V_2(q;p) &= pq(2) + p(1-q)(1) + (1-p)q(0) + (1-p)(1-q)(3) \\ &= 4pq - 2p - 3q + 3. \end{aligned}$$

Given player 2's mixed strategy $\sigma_2 = (q, 1-q)$, we determine player 1's best response $\sigma_1^* = (p^*, 1-p^*)$. Note that

$$\frac{\partial V_1(p;q)}{\partial p} = 5q - 2 \implies \frac{\partial V_1(p;q)}{\partial p} \begin{cases} > 0, & \text{if } q > \frac{2}{5}, \\ = 0, & \text{if } q = \frac{2}{5}, \\ < 0, & \text{if } q < \frac{2}{5}. \end{cases}$$

Thus player 1's best response is

$$p^* = \begin{cases} 1, & \text{if } q > \frac{2}{5}, \\ x, & \text{if } q = \frac{2}{5}, \\ 0, & \text{if } q < \frac{2}{5}, \end{cases}$$

where x is any number between 0 and 1. (When $q = \frac{2}{5}$, the payoff $V_1\big(p; \frac{2}{5}\big) = \frac{6}{5}$ is independent of p.)

Given player 1's mixed strategy $\sigma_1 = (p, 1-p)$, we determine player 2's best response $\sigma_2^* = (q^*, 1-q^*)$. Note that

$$\frac{\partial V_2(q;p)}{\partial q} = 4p - 3 \implies \frac{\partial V_2(q;p)}{\partial q} \begin{cases} > 0, & \text{if } p > \frac{3}{4}, \\ = 0, & \text{if } p = \frac{3}{4}, \\ < 0, & \text{if } p < \frac{3}{4}. \end{cases}$$

Thus player 2's best response is

$$q^* = \begin{cases} 1, & \text{if } p > \frac{3}{4}, \\ y, & \text{if } p = \frac{3}{4}, \\ 0, & \text{if } p < \frac{3}{4}, \end{cases}$$

where y is any number between 0 and 1. (When $p = \frac{3}{4}$, the payoff $V_2\big(q; \frac{3}{4}\big) = \frac{3}{2}$ is independent of q.)

The two curves p^* and q^* intersect at $(p^*, q^*) = (0,0)$, $\big(\frac{3}{4}, \frac{2}{5}\big)$ and $(1,1)$. The first and third intersection points correspond to the pure-strategy Nash equilibria (B,B) and (S,S) respectively. The unique (non-degenerate) mixed-strategy Nash equilibrium is

$$\sigma_1^* = (p^*, 1-p^*) = \left(\frac{3}{4}, \frac{1}{4}\right), \qquad \sigma_2^* = (q^*, 1-q^*) = \left(\frac{2}{5}, \frac{3}{5}\right).$$

1.3.2 Indifference principle

The indifference principle states that given a mixed (or pure) strategy adopted by other players, the player i's payoff remains constant over all pure strategy s_i chosen by the player in her mixed strategy.

Theorem (Indifference principle)**.** Let $\sigma^* = (\sigma_1^*, \sigma_2^*, \ldots, \sigma_n^*)$ be a mixed-strategy Nash equilibrium. For any pair of actions $a_i^1 \in S_i$ and $a_i^2 \in S_i$ with $\sigma_i^*(a_i^1) > 0$ and $\sigma_i^*(a_i^2) > 0$, we have

$$V_i(a_i^1; \sigma_{-i}^*) = V_i(a_i^2; \sigma_{-i}^*).$$

e.g.

	Work (W)	Do nothing (NW)
Aid (A)	$(3,2)$	$(-1,3)$
Does not aid (NA)	$(1,1)$	$(0,0)$

Identify all Nash equilibria using the indifference principle.
Solution: We determine the best response of each player. This can be done using the indifference principle. For the government (Player 1), let $\sigma_2 = (q, 1-q)$ be a given mixed strategy of Player 2 and let $\sigma_1 = (p, 1-p)$ be Player 1's mixed strategy. Consider

$$V_1(A; \sigma_2) = V_1(NA; \sigma_2) \iff 3q + (-1)(1-q) = q + 0(1-q) \iff q = \frac{1}{3}.$$

Now we examine three cases:

- If $q = \frac{1}{3}$, the indifference principle implies that Player 1's best response is to mix between A and NA (i.e. $p \in (0,1)$). The expected payoff of Player 1 is

$$V_1(\sigma_1; \sigma_2) = pV_1(A; \sigma_2) + (1-p)V_1(NA; \sigma_2) \overset{V_1(A;\sigma_2)=V_1(NA;\sigma_2)}{=} V_1(A; \sigma_2),$$

  which is independent of p. Hence the best response is $\sigma_1^* = (p, 1-p)$ for any $p \in [0,1]$.

- If $q < \frac{1}{3}$, we have $V_1(A; \sigma_2) < V_1(NA; \sigma_2)$. By the proof of the indifference principle, Player 1 should choose NA with certainty. Thus the best response is $\sigma_1^* = (0,1)$ (i.e. $p = 0$).

- If $q > \frac{1}{3}$, we have $V_1(A; \sigma_2) > V_1(NA; \sigma_2)$. Similarly, Player 1 should choose A with certainty, so the best response is $\sigma_1^* = (1,0)$ (i.e. $p = 1$).

Next we find Player 2's best response σ_2^* given Player 1's strategy $\sigma_1 = (p, 1-p)$. First,

$$V_2(W; \sigma_1) = V_2(NW; \sigma_1) \iff 2p + 1(1-p) = 3p + 0(1-p) \iff p = \frac{1}{2}.$$

Again we consider three cases:

- If $p = \frac{1}{2}$, the indifference principle implies that Player 2's best response is to mix between W and NW (i.e. $q \in (0,1)$). The expected payoff of Player 2 is

$$V_2(\sigma_2; \sigma_1) = qV_2(W; \sigma_1) + (1-q)V_2(NW; \sigma_1) \overset{V_2(W;\sigma_1)=V_2(NW;\sigma_1)}{=} V_2(W; \sigma_1),$$

  which is independent of q. Hence the best response is $\sigma_2^* = (q, 1-q)$ for any $q \in [0,1]$.

- If $p < \frac{1}{2}$, we have $V_2(W;\sigma_1) > V_2(NW;\sigma_1)$. By the indifference principle, Player 2 should choose W with certainty, so the best response is $\sigma_2^* = (1,0)$ (i.e. $q = 1$).
- If $p > \frac{1}{2}$, we have $V_2(W;\sigma_1) < V_2(NW;\sigma_1)$. Similarly, Player 2 should choose NW with certainty, so the best response is $\sigma_2^* = (0,1)$ (i.e. $q = 0$).

Thus, the unique (non-degenerate) mixed-strategy Nash equilibrium is

$$\sigma_1^* = (p^*, 1-p^*) = \Big(\frac{1}{2}, \frac{1}{2}\Big), \qquad \sigma_2^* = (q^*, 1-q^*) = \Big(\frac{1}{3}, \frac{2}{3}\Big).$$

Remark:
1. We can also claim that a game has no mixed strategy Nash equilibrium if the p^* or q^* is not in $(0,1)$.
2. And if you have 3 pure strategy to choose for 2 players respectively(e.g. Paper-Scissor-Rock), then you can first examine if it is possible for one player to use pure strategy, then examine if it is possible for one player to use mix only 2 pure strategies.

Proof of dominated strategy never play mixed strategy Nash equilibrium:
In an n-person game, let $s_i^0 \in S_i$ be a strictly dominated pure strategy of player i. Show that player i never plays s_i^0 in any mixed-strategy Nash equilibrium $\sigma^* = (\sigma_1^*, \sigma_2^*, \ldots, \sigma_n^*)$. Suppose $\sigma_i^*(s_i^0) > 0$.

Since s_i^0 is strictly dominated, there exists another strategy s_i' that strictly dominates it; i.e.

$$V_i(s_i^0; s_{-i}) < V_i(s_i'; s_{-i}) \qquad \text{for every pure strategy } s_{-i} \in S_{-i}.$$

Construct the following mixed strategy for player i:

$$\sigma_i'(s_i) = \begin{cases} \sigma_i^*(s_i) & \text{if } s_i \neq s_i^0,\ s_i', \\ 0 & \text{if } s_i = s_i^0, \\ \sigma_i^*(s_i') + \sigma_i^*(s_i^0) & \text{if } s_i = s_i'. \end{cases}$$

By definition, we can show $V_i(\sigma_i'; \sigma_{-i}^*) > V_i(\sigma_i; \sigma_{-i}^*)$. Thus σ_i^* is not the best response to σ_{-i}^*, so we can conclude that $\sigma_i^*(s_i^0) = 0$.

1.3.3 Dominated strategy: Extended definition

Definition (Dominated strategy: Extended definition)**.** A pure strategy $s_i \in S_i$ is said to be strictly dominated by a mixed strategy σ_i if and only if

$$V_i(s_i; s_{-i}) < V_i(\sigma_i; s_{-i}) \ldots (*)$$

for any combination of opponents' pure strategies $s_{-i} \in S_{-i}$.
A pure strategy $s_i \in S_i$ is said to be a dominated strategy if it is strictly dominated by some mixed strategy σ_i.

Remarks about the definition

- The definition above is stated for the case where the rival uses a *pure* strategy. However, one can show that for any mixed strategy σ_{-i} of the rival,

$$V_i(s_i; \underline{\sigma_{-i}}) = \sum_{s_{-i} \in S_{-i}} \sigma_{-i}(s_{-i}) V_i(s_i; s_{-i}) < \sum_{s_{-i} \in S_{-i}} \sigma_{-i}(s_{-i}) V_i(\sigma_i; s_{-i}) = V_i(\sigma_i; \underline{\sigma_{-i}}).$$

Thus, the dominance relation remains valid even when the rival employs a mixed strategy. In practice, checking inequality $(*)$ is often the most convenient way to identify dominated strategies.

- If a pure strategy s_i is dominated by a mixed strategy σ_i, one can show that s_i can be dominated by another mixed strategy σ_i' with $\sigma_i'(s_i) = 0$.

  To see this, note that

$$V_i(s_i; s_{-i}) < V_i(\sigma_i; s_{-i}) = \sum_{s \in S_i} \sigma_i(s) V(s; s_{-i}) = \sum_{s \neq s_i} \sigma_i(s) V(s; s_{-i}) \; + \; \sigma_i(s_i) V(s_i; s_{-i}).$$

$$\implies V_i(s_i; s_{-i}) < \sum_{s \neq s_i} \underbrace{\frac{\sigma_i(s)}{1 - \sigma_i(s_i)}}_{\sigma_i'(s)} V(s; s_{-i}),$$

  Consider a mixed strategy σ_i' defined by

$$\sigma_i'(s_i) = 0 \quad \text{and} \quad \sigma_i'(s) = \frac{\sigma_i(s)}{1 - \sigma_i(s_i)} \text{ for } s \neq s_i.$$

  One can verify that

$$\sum_{s \in S_i} \sigma_i'(s) = \sum_{s \neq s_i} \frac{\sigma_i(s)}{1 - \sigma_i(s_i)} = \frac{1 - \sigma_i(s_i)}{1 - \sigma_i(s_i)} = 1,$$

  and

$$V_i(s_i; s_{-i}) < \sum_{s \neq s_i} \underbrace{\frac{\sigma_i(s)}{1 - \sigma_i(s_i)}}_{\sigma_i'(s)} V_i(s; s_{-i}) = V_i(\sigma_i'; s_{-i}).$$

Theorem (Nash's existence theorem)**.** For any n-person normal form games with finite strategic sets S_i for every player, there exists at least one mixed strategy Nash equilibrium.

(*Note: Here, pure strategy Nash equilibrium is treated as a special case of mixed strategy Nash equilibrium)
Proof: Given the strategy set S_i of each player, we define ΔS_i as the collection of all mixed strategies available to player i; that is,

$$\Delta S_i = \Big\{ \underbrace{(\sigma_i(a_{i1}), \sigma_i(a_{i2}), \ldots, \sigma_i(a_{iN_i}))}_{\sigma_i} \;\Big|\; \sigma_i(a_{i1}) + \cdots + \sigma_i(a_{iN_i}) = 1 \Big\}.$$

Let $\Delta S = \Delta S_1 \times \Delta S_2 \times \cdots \times \Delta S_n$ be the set of mixed-strategy profiles of the n players. Given a strategy profile $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_n)$, recall that $BR_i(\sigma)$ denotes player i's best response (a mixed strategy) to the rivals' strategies σ_{-i}, and $BR(\sigma) = \big(BR_1(\sigma), BR_2(\sigma), \ldots, BR_n(\sigma)\big)$ denotes the best response of n players to the strategic profile σ.
Recall that a profile $\sigma^* = (\sigma_1^*, \sigma_2^*, \ldots, \sigma_n^*)$ is a Nash equilibrium if and only if σ_i^* is a best response to σ_{-i}^* for every player i; that is, $BR_i(\sigma^*) = \sigma_i^*$. Hence

$$BR(\sigma^*) = \sigma^*.$$

Thus σ^* is a fixed point of the correspondence $BR(\sigma)$. Consequently, proving the existence of a Nash equilibrium reduces to showing that the correspondence $BR : \Delta S \rightrightarrows \Delta S$ has a fixed point.
To see if this fixed point exists, we introduce the following theorem:

Theorem (Kakutani's fixed point theorem). Let $C : X \rightrightarrows X$ be a correspondence where $X \subseteq \mathbb{R}^p$ and $C(x) \subseteq X$ for all $x \in X$. Suppose that

1. X is non-empty, closed, bounded, and convex;
   (A set X is said to be closed if and only if for any sequence $\{x_n\}_{n\in\mathbb{N}}$ with $x_n \in X$ and $\lim_{n\to\infty} x_n = x_0$, we have $x_0 \in X$.)
2. $C(x)$ is non-empty for all $x \in X$;
   (A set B is said to be convex if and only if for any $x, y \in B$, we have $\alpha x + (1-\alpha)y \in B$ for any $\alpha \in [0,1]$.)
3. $C(x)$ is convex for all x;
4. C has closed graph, i.e. the set $\{(x, C(x)) : x \in X\}$ is closed.

Then there exists $x^* \in X$ such that
$$C(x^*) = x^*.$$

Thus, let $p_{ij} = \sigma_i(a_{ij})$, where $a_{ij} \in S_i$.

For condition 1:
Obviously ΔS_i is non-empty for all i, so ΔS is non-empty.
$0 \le p_{ij} \le 1$ for all i, j, thus $\Delta S_i \subseteq [0,1]^{N_i}$ and ΔS_i is bounded.
let $\sigma_i^{(n)} \in \Delta S_i$ (where $\sigma_i^{(n)} = (p_{i1}^{(n)}, p_{i2}^{(n)}, \ldots, p_{iN_i}^{(n)})$) be a sequence of vectors which $\lim_{n\to\infty} \sigma_i^{(n)} = \sigma_i^* = (p_{i1}^*, p_{i2}^*, \ldots, p_{iN_i}^*)$. $p_{ij}^{(n)} \ge 0$ for all n, thus $p_{ij}^* = \lim_{n\to\infty} p_{ij}^{(n)} \ge 0$.
$\sigma_i^{(n)} \in \Delta S_i$, so that $p_{i1}^{(n)} + p_{i2}^{(n)} + \cdots + p_{iN_i}^{(n)} = 1$. Taking limit on both sides, we have $p_{i1}^* + p_{i2}^* + \cdots + p_{iN_i}^* = 1$.
It follows that $\sigma_i^* \in \Delta S_i$. So that ΔS_i is closed.
Thus, for any $\sigma_i^{(1)}, \sigma_i^{(2)} \in \Delta S_i$ and $\alpha \in [0,1]$, we let $\sigma = \alpha\sigma_i^{(1)} + (1-\alpha)\sigma_i^{(2)}$. One can verify that:
1. $\alpha p_{ij}^{(1)} + (1-\alpha)p_{ij}^{(2)} \ge 0$ as $p_{ij}^{(1)}, p_{ij}^{(2)} \ge 0$
2. $\displaystyle\sum_{j=1}^{N_i}\left(\alpha p_{ij}^{(1)} + (1-\alpha)p_{ij}^{(2)}\right) = \alpha \underbrace{\sum_{j=1}^{N_i} p_{ij}^{(1)}}_{=1} + (1-\alpha)\underbrace{\sum_{j=1}^{N_i} p_{ij}^{(2)}}_{=1} = 1.$

So $\sigma \in \Delta S_i$ and ΔS_i is convex.

For condition 2:
Note that the (expected) payoff function $V_i(\sigma_i; \sigma_{-i}) = \sum p_{ij} V_i(a_{ij}; \sigma_{-i})$ is continuous with respect to p_{ij} (as it is linear in p_{ij}) and the domain ΔS_i is closed and bounded. Thus from extreme value theorem, function $V_i(\sigma_i; \sigma_{-i})$ has a maximum in ΔS_i, which means best response exists.
Thus $BR_i(\sigma)$ is non-empty.

For condition 3:
let σ_i^* and σ_i^{**} be two best responses to the rival's strategy σ_{-i}, and let $M^* = V_i(\sigma_i^*; \sigma_{-i}) = V_i(\sigma_i^{**}; \sigma_{-i})$.
As σ_i^* is a best response, we have $M^* = V_i(\sigma_i^*; \sigma_{-i}) \ge V_i(\sigma_i; \sigma_{-i})$ for all $\sigma_i \in \Delta S_i$.
Then we consider the strategy $\sigma_0 = \beta\sigma_i^* + (1-\beta)\sigma_i^{**}$, its payoff function is:

$$V_i(\sigma_0; \sigma_{-i}) = \sum_{j=1}^{N_i}\left(\beta p_{ij}^* + (1-\beta)p_{ij}^{**}\right)V_i(a_{ij}; \sigma_{-i}) = \beta \underbrace{\sum_{j=1}^{N_i} p_{ij}^* V_i(a_{ij}; \sigma_{-i})}_{=M^*} + (1-\beta)\underbrace{\sum_{j=1}^{N_i} p_{ij}^{**} V_i(a_{ij}; \sigma_{-i})}_{=M^*} = M^* \ge V_i(\sigma_i; \sigma_{-i})$$

for all $\sigma_i \in \Delta S_i$. Thus σ_0 is also a best response and $\sigma_0 \in BR(\sigma)$, thus $BR(\sigma)$ is convex.

For condition 4:
let $\lim_{n\to\infty}(\sigma^{(n)}, BR(\sigma^{(n)})) = (\sigma^*, \sigma^{**})$. To prove the graph is closed, it remains to show $\sigma^{**} = BR(\sigma^*)$, i.e. $\sigma^{**} = \lim_{n\to\infty} BR(\sigma^{(n)})$ is the best response to σ^*.
$BR(\sigma^{(n)})$ is a best response to $\sigma^{(n)}$, thus

$$V_i\big(BR_i(\sigma^{(n)}); \sigma_{-i}^{(n)}\big) \geq V_i\big(\sigma_i; \sigma_{-i}^{(n)}\big) \quad \ldots\ldots(*)$$

for any σ_i.
$V_i(\sigma_i; \sigma_{-i}) = \sum\limits_{a_i \in S_i} \sum\limits_{a_{-i} \in S_{-i}} \sigma_i(a_i)\sigma_{-i}(a_{-i})V_i(a_i; a_{-i})$ is a continuous function of $\sigma_i(s_i)$ and $\sigma_{-i}(s_{-i})$ (and hence of σ_i and σ_{-i}). Taking $n \to \infty$ in $(*)$, we know that

$$\lim_{n\to\infty} V_i\big(BR_i(\sigma^{(n)}); \sigma_{-i}^{(n)}\big) \geq \lim_{n\to\infty} V_i\big(\sigma_i; \sigma_{-i}^{(n)}\big)$$

$$\Rightarrow V_i(\sigma_i^{**}; \sigma_{-i}^*) \geq V_i(\sigma_i; \sigma_{-i}^*).$$

Then we can conclude that σ_i^{**} is the best response to σ_{-i}^* and $\sigma^{**} = BR(\sigma^*)$.

Thus, from Kakutani's fixed point theorem, there exists σ^* s.t. $BR(\sigma^*) = \sigma^*$ and the mixed strategy Nash equilibrium exists.

1.4 Additional topic: Existence of pure strategy Nash equilibrium in 2-person finite games

We focus on 2-person zero-sum games(e.g. Paper-Scissor-Rock), that is,

$$V_1(s_1; s_2) + V_2(s_2; s_1) = 0$$

for any $s_1, s_2 \in S_1 \times S_2$.
1)
For player1: since all players are rational, so player 2 would try to maximized her payoff, thus payoff choosing strategy $s_1 = \underset{s_2}{min} V_1(s_1; s_2)$, and since player1 want to maximize her payoff, so the payoff of player1 =

$$v^- = \underset{s_1}{max}(\underset{s_2}{min} V_1(s_1; s_2))$$

v^- is called lower value of the games which indicates the minimum guarantee cutoff that player 1 can receive from the games.
For player2, similarly, she would expect the payoff of choosing strategy $s_1 = -\underset{s_1}{max} V_1(s_1; s_2)$, and the player2's expected payoff $= -\underset{s_2}{min}(\underset{s_1}{max} V_1(s_1; s_2))$, thus

$$v^+ = \underset{s_2}{min}(\underset{s_1}{max} V_1(s_1; s_2))$$

is called upper value of the games which indicates the maximum loss that player 2 can suffer from the games.
Now consider Nash equilibrium (s_1^*, s_2^*).
For player1,

$$V_1(s_1^*; s_2^*) = \underset{s_2}{min} V_1(s_1^*; s_2) \leq \underset{s_1}{max}(\underset{s_2}{min} V_1(s_1; s_2)) = v^-$$

also with

$$V_1(s_1^*; s_2^*) = \underset{s_1}{max} V_1(s_1^*; s_2) \geq \underset{s_1}{max}(\underset{s_2}{min} V_1(s_1; s_2)) = v^-$$

Thus we have

$$V_1(s_1^*; s_2^*) = v^-$$

For player2, similarly, $V_2(s_2^*; s_1^*) = \underset{s_2}{max}(\underset{s_1}{min} V_2(s_2; s_1))$, thus

$$-V_1(s_1^*; s_2^*) = \underset{s_2}{max}(-\underset{s_1}{min} V_1(s_1; s_2)) = -\underset{s_2}{min}(\underset{s_1}{max} V_1(s_1; s_2))$$

Thus

$$V_1(s_1^*; s_2^*) = \underset{s_2}{min}(\underset{s_1}{max} V_1(s_1; s_2)) = v^+$$

Combining the result, we deduce that $\mathbf{v^- = V_1(s_1^*; s_2^*) = v^+}$ if $\mathbf{(s_1^*, s_2^*)}$ is Nash equilibrium.

2)

let s_1^* be a strategy such that $\underset{s_2}{min} V_1(s_1^*; s_2) = v^-$. On the other hand, we let s_2^* be a strategy such that $\underset{s_1}{max} V_1(s_1; s_2^*) = v^+$.

Consider player1:

$$V_1(s_1^*; s_2^*) \geq \underset{s_2}{min} V_1(s_1^*; s_2) = v^- = v^+ = \underset{s_1}{max} V_1(s_1; s_2^*) \geq V_1(s_1; s_2^*)$$

for any $s_1 \in S_1$. This implies that s_1^* is the player 1's best response to s_2^*.

Consider player2: similarly,

$$V_2(s_2^*; s_1^*) = -V_1(s_1^*; s_2^*) = -\max_{s_1} V_1(s_1; s_2^*) = -v^+ = -v^- = -\min_{s_2} V_1(s_1^*; s_2) \geq -V_1(s_1^*; s_2) = V_2(s_2; s_1^*).$$

for any $s_2 \in S_2$. This implies that s_2^* is the player 2's best response to s_1^*.

Hence, we conclude that (s_1^*, s_2^*) is the desired pure strategy Nash equilibrium.

Theorem (Existence of pure strategy Nash equilibrium in 2-person finite games)**.** A two-person finite games has the pure strategy Nash equilibrium if and only if

$$v^- = \max_{s_1}(\min_{s_2} V_1(s_1; s_2)) = \min_{s_2}(\max_{s_1} V_1(s_1; s_2)) = v^+$$